Heartwise / Cardio Signal

Web-based heart disease screening system

Project overview

Heartwise is an end-to-end web application that transforms a trained decision-tree classifier into a simple, explainable heart-disease screening tool. It accepts ten routine clinical inputs, applies an embedded model, and displays a screening estimate without claiming a diagnosis.

The project is built with ASP.NET Core Razor Pages and a shallow decision tree exported to a compact JSON model, keeping inference transparent, lightweight, and independent of Python at runtime.

Methodology

The solution uses the UCI heart disease dataset, leakage-aware preprocessing, and tuned decision-tree selection. The final model is embedded into the app so users can interact with a reproducible ML pipeline from the browser.

System goals

This application is designed for learning, demonstration, and small-scale research. It keeps model logic inspectable, avoids a Python runtime in production, and stores results in MySQL for review and traceability.

10	21	MySQL
INPUTS	TREE NODES	STORAGE

Problem statement

Cardiovascular disease remains a leading global health burden, yet many model demos are hidden in notebooks or inaccessible dashboards. This project packages a transparent risk model into a browser-based tool that is easier to read, run, and explain.

Model inputs

Age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, and ST slope.

System architecture

? Data preparation and preprocessing

? Model selection and smoothing

? JSON export of the tree

? ASP.NET Core web interface

? MySQL persistence for saved screenings

Objectives

? Build a reproducible pipeline from CSV to web app

? Use a simple, inspectable tree model

? Keep runtime independent of Python

? Persist screening results in MySQL

? Evaluate the model honestly and document limitations

Key results

Selected model: entropy-based decision tree, depth 4, 21 nodes, 11 leaves

Accuracy: 0.822 | ROC AUC: 0.893

Log loss: 0.399 | Brier score: 0.124

The output is an educational screening estimate, not a diagnosis.

Limitations

This is an educational prototype, not a clinical device. The model requires external validation and fairness review before any medical deployment.